Certificates as Constraints: A Certified Reference Monitor for Learned Task Allocation in UAV Swarms under Navigation Attacks

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***Abstract*—Learned controllers are entering unmanned aerial vehicle (UAV) fleets, and so are the attacks that degrade navigation. A jammer forces aircraft to dead-reckon, an undetected relay-delay attack makes a stale correction arrive all at once, and a compromised aircraft can misreport its own state. Recent work bounds the resulting position error of each aircraft by a formal, time-varying certificate $\rho_i(t)$. We ask what a fleet can safely do with such certificates when its task allocator is a learned policy that may itself be poorly trained, poisoned or adversarially driven. A task with alert limit α_j may be held only while $\rho_i(t) \le \alpha_j$, so the feasible set moves with the attack. We place a certified projection layer between the policy and the fleet, acting as a reference monitor. For *every* policy parameter and every certificate trajectory the adversary can produce, it executes no infeasible assignment, and it covers the maximum number of tasks the certificates allow. Because a scalar certificate makes eligibility nested, matching theory yields exact, linear-time security monitors: certified coverage, the number of certificate collapses the fleet tolerates, and the exact worst case against an adversary who degrades any b aircraft. A predictive rule confines inter-epoch exposure to certificate jumps and fixes the allocator rate; with measured constants it is 5.01 Hz. Misreporting aircraft shift the coverage monitor by at most their number and are the only aircraft that can violate the constraint. A population lift turns these per-operation guarantees into fleet-sizing statements and exposes their independence obligation: common-cause jamming breaks it, and the independent model then understates the probability of losing six of eight tasks by $3.7\times$. Every constant comes from released measurements, and a brute-force oracle confirms every closed form on 83,872 checks with 0 disagreements. In a multi-agent study up to 96 aircraft, the monitor executes no infeasible assignment during learning, where in-expectation baselines (PPO-, PID- and MAPPO-Lagrangian, QMIX) execute millions, and it lifts every learned policy to maximal coverage at deployment. A classical consensus auction under the same monitor outperforms every learned allocator: the security guarantee comes from the monitor, not from learning.**

I. Introduction

A swarm of UAVs flies a monitoring or delivery mission over a city. Each task tolerates only a bounded position error: a 2 m corridor between buildings, a 20 m transect over open ground. The navigation of every aircraft is under attack. GNSS jamming and spoofing are practical and cheap [1]–[3]. A recent line of work bounds how far an aircraft can drift when an adversary delays its navigation input just below an intrusion detector's threshold, and turns the bound into a per-aircraft certificate $\rho_i(t)$ [4]. Its cryptographic cost is quantified in [5], and its relation to aviation safety objectives in [6]. The certificate is not a constant: it grows while a jammer forces the aircraft to dead-reckon, and it jumps when a compromised relay delivers a stale correction.

Meanwhile, the decision of *which aircraft serves which task* is increasingly made by a learned multi-agent policy. Learned policies are a security liability of their own. They can be driven into unsafe actions by adversarial observations [7], by adversarial co-players [8] or by backdoors planted in training [9], [10]. Safe reinforcement learning (RL) offers two kinds of guarantee, and neither fits. Lagrangian and trust-region methods for constrained Markov decision processes (CMDPs) [11]–[14] satisfy the constraint *in expectation*, and only for the policy that training produced. A fleet that flies outside a certified envelope on 5% of steps is not acceptable, and a poisoned policy voids the expectation anyway. Shields, safety layers and control barrier functions [15]–[18] hold a constraint at every step, but against a safe set fixed in advance, and they guarantee safety, not the mission.

We take the security engineer's view. The allocator is untrusted; the certificate is the security policy; and the mechanism that enforces it must be a small, always-invoked, verifiable reference monitor [19], in the spirit of runtime assurance for aircraft [20], [21]. The question becomes: *given certificates that an adversary moves, which assignments can be executed safely, how much of the mission survives, and how much can a learned policy add on top?*

a) What is and is not new.: The combinatorial core is classical. With a scalar certificate, eligibility has nested neighbourhoods: an aircraft that qualifies for a demanding task qualifies for every easier one. On such *difference graphs* [22], Ore's deficiency theorem [23] gives the maximum coverage in closed form, the condition for surviving b removals is the b-extended Hall condition of robust assignment [24], and pointwise feasibility under projection is the standard shield guarantee [15], [25]. We use these as tools, with attribution. The contribution is their composition with a certificate that an adversary moves, against a learned allocator that we do not trust, and the security results that composition makes possible.

b) Contributions.:

1) **Threat model and security goals** (Section II-C). We model four adversaries: a jammer, a relay-delay attacker, misreporting aircraft and a compromised policy. We state integrity, availability and bounded-influence goals, and we grade the guarantee by quantifier against safe RL (Table I).
2) **A certified reference monitor** (Theorem 1). Placed in the environment, the projection makes every executed assignment feasible at every decision epoch for every policy parameter, poisoned or not. It also assigns the maximum number of tasks, then maximises value, then follows the policy. Policy gradients stay unbiased for the projected process, so the monitor can be present while the policy learns.
3) **Exact security monitors** (Section III): worst-case coverage after any b certificate collapses and after graded degradation of any b aircraft, both exact and attained by an adversary (Propositions 1 and 2); a hysteresis bound (Theorem 2); a predictive rule under which exposure between decisions comes only from certificate jumps, with the allocator rate it implies (Theorem 3); and a misreporting bound (Proposition 3), with decentralised agreement on the monitor's inputs by resilient consensus.
4) **A population lift with explicit obligations** (Proposition 4) that turns the per-operation guarantee into fleet-sizing statements and shows quantitatively how common-cause jamming breaks it.
5) **A measured, reproducible evaluation.** Every constant is derived from released flight and network data, and a brute-force oracle verifies every closed form. A multi-agent RL (MARL) study with stronger in-expectation baselines and larger fleets, a scalability study, a consensus study with Byzantine aircraft and a vector-certificate study answer seven research questions. They separate what a learned policy contributes from what the monitor enforces, and show that a classical auction under the monitor is the stronger allocator in our setting.

II. Background, Threat Model and Goals

A. The certified protection level

The parent certificate [4] considers an adversary that delays relayed navigation messages while staying below a detector's threshold. Let $\Delta_i(t)$ be the *staleness* of aircraft i's navigation input: the age of the newest trustworthy fix it is acting on. Staleness grows at one second per second while the aircraft dead-reckons, and it jumps by the hidden per-hop residual when a compromised relay delivers a delayed correction. Against this adversary, the certificate bounds the position error by a Grönwall tube,

$$\|\hat{p}_i(t) - p_i(t)\| \;\le\; \rho_i(t) \;\coloneqq\; e^{LT}\,\gamma_{\sup}\,\Delta_i(t), \tag{1}$$

where $\gamma_{\sup}$ is the supremum rate at which staleness becomes position error and e^{LT} amplifies it through closed-loop dynamics with local Lipschitz constant L over a horizon T. All three factors are measured in the released artifact: $\gamma_{\sup} = 1.625\,\text{m/s}$ over 674 post-onset samples of real GPS jamming and spoofing flights; $L = 1.181$ with $T = 1\,\text{s}$; benign residuals no larger than 0.178 s, and undetected malicious residuals with median 3.94 s and maximum 4.84 s, over 15,392 relay hops. Writing $L_\rho \coloneqq e^{LT}\gamma_{\sup} = 5.30\,\text{m/s}$, a dead-reckoning aircraft's protection level grows at most at rate L_ρ. A single undetected two-hop relay attack moves it by about 41.8 m, past every alert limit we use. The benign floor is $\rho_0 = L_\rho \cdot 0.178 \approx 0.94\,\text{m}$.

We borrow two terms from GNSS integrity monitoring. The *protection level* is a runtime bound on position error, and the *alert limit* is the largest error an operation tolerates; an operation is unavailable whenever the first exceeds the second [26]. We call $\rho_i(t)$ the *certified protection level*. Each task j carries an alert limit $\alpha_j > 0$, its corridor half-width ($\{2,5,10,20\}$ m in our missions). Unlike a receiver protection level, which overbounds error at a target integrity risk, (1) is a worst-case bound conditional on stated assumptions [4]; Section III-D returns to that difference.

B. System model

A fleet $\mathcal{N} = \{1,\dots,N\}$ serves tasks $\mathcal{T} = \{1,\dots,M\}$, each with a site, an alert limit α_j and a value $v_j > 0$. Decisions are taken at epochs $t_k = k\Delta t$. An *assignment* is a map $\sigma : \mathcal{N} \to \mathcal{T} \cup \{\varnothing\}$ that is injective on $\mathcal{T}$: a matching in $\mathcal{N} \times \mathcal{T}$. At each epoch every aircraft reports $\rho_i(t_k)$, signed with its own key (the signing cost is budgeted in [5]); a learned policy π_θ proposes an assignment from its observations; and the *allocation layer* decides what is executed. The allocation layer runs in a trusted component, a ground station or a replicated flight-management function; it is the only path from a proposal to a flight command.

C. Threat model

The adversary may combine the following capabilities, adaptively and with full knowledge of the allocator and the policy.

A1 *Jamming and sensor denial.* Jammers, possibly mobile and covering several aircraft at once, deny GNSS fixes. Affected aircraft dead-reckon, and their certificate grows at rate at most L_ρ.

A2 *Relay delay.* The adversary controls up to $H = 2$ relay hops and delays corrections below the detector's threshold [4]. Certificates jump upward by the hidden residual, without warning.

A3 *Compromised aircraft.* Up to f aircraft are Byzantine: they report false certificates, tell different neighbours different values, drop the reports they should relay, and send arbitrary values in any agreement protocol.

A4 *Compromised policy.* The policy parameters θ are arbitrary: undertrained, backdoored [9], or steered by adversarial observations [7].

A1 and A2 together let the adversary choose any certificate trajectory that the parent certificate admits. We therefore model each $\rho_i(\cdot)$ as an arbitrary nonnegative, bounded càdlàg process (right-continuous with left limits); a jammer is not a distribution.

Assumption 1 (Trust base)**.** (a) *Certificate soundness* (from [4]): whenever an honest aircraft reports $\rho_i(t)$, $\|\hat{p}_i(t) - p_i(t)\| \le \rho_i(t)$. (b) The signing keys of honest aircraft are not compromised, so a relay cannot alter another aircraft's report. (c) The allocation layer executes as specified. Only Theorem 1(iii) and the learning study additionally assume a Markov threat model.

Out of scope. Attacks that break the certificate itself are the parent paper's concern; physical attacks, collision avoidance, and denial of the command link beyond dropping relayed reports are not considered.

D. Security goals

Constraint (C) is the security policy: an aircraft may hold a task only while its certificate is within the task's alert limit,

$$\text{(C)} \quad \rho_i(t_k) \le \alpha_{\sigma_{t_k}(i)} \quad \text{for every } i \text{ with } \sigma_{t_k}(i) \neq \varnothing, \text{ every } k. \tag{2}$$

G1 *Integrity.* (C) holds at every epoch of training and deployment, for every θ (A4) and every certificate trajectory (A1, A2).

G2 *Availability under attack.* Integrity must not be bought by idling: the executed assignment covers as many tasks as the certificates allow, and the worst case after the adversary degrades any b aircraft is known exactly and in real time.

G3 *Bounded exposure between decisions.* Between epochs, a holder becomes infeasible only through a certificate jump, which no rate bound can anticipate.

G4 *Bounded influence of liars.* f Byzantine aircraft (A3) shift every monitored quantity by at most f, and only they can violate (C).

E. The fleet CMDP and its quantifier

The fleet is a CMDP [11] whose state s_t contains positions, task sites and every $\rho_i(t)$. Its action is an assignment, and its reward is *certified coverage*, the value of tasks whose holder is on site and certified: $r_{\text{cert}}(t) = \sum_j v_j \mathbb{1}[\text{holder of } j \text{ on site and } \rho_{\sigma^{-1}(j)}(t) \le \alpha_j]/\sum_j v_j$. The constraint is (C), pointwise. The *eligibility graph* $G(t)$ joins i to j iff $\rho_i(t) \le \alpha_j$, and (C) says that executed assignments are matchings of $G(t_k)$. Sort the alert limits ascending, $\alpha_{(1)} \le \cdots \le \alpha_{(M)}$, and define the *count profile* $n_k(t) := |\{i : \rho_i(t) \le \alpha_{(k)}\}|$. Since $\rho_i \le \alpha_{(k)}$ implies $\rho_i \le \alpha_{(k')}$ for all $k' \ge k$, neighbourhoods in $G(t)$ are nested: $G(t)$ is a *difference* (chain) graph [22], [27]. Table I places the guarantee among those of safe RL.

F. Problem statement

Given an untrusted policy class that proposes assignments, we seek: **P1** an allocation mechanism that meets G1 and G2 for any policy parameters (leaving every aircraft idle meets G1 trivially, so coverage must be part of the guarantee); **P2** an exact, real-time characterisation of the coverage the fleet can certify, and of how it degrades under attack; **P3** rules that meet G3 and avoid chattering as $\rho_i(t)$ fluctuates, and a mechanism for G4 without a trusted aggregator; **P4** a sound route from the per-operation guarantee ($\mathbf{Q}_{\text{op}}$) to the per-population statements that fleet sizing and airworthiness objectives require ($\mathbf{Q}_{\text{pop}}$), with every assumption explicit. Figure 1 summarises the approach. Section V answers seven research questions: **RQ1** violations during learning; **RQ2** coverage under degradation, against the bounds; **RQ3** what the policy adds over the monitor, against stronger baselines and at larger fleet sizes; **RQ4** scalability; **RQ5** dynamics between decision epochs; **RQ6** the population lift; **RQ7** Byzantine aircraft and decentralised agreement.

III. Design: Certified Allocation under Nested Constraints

Proofs are in Appendix A. Throughout, $d(t)$, $n_k(t)$ and $G(t)$ are evaluated at the current certificates. We drop t when no confusion arises.

A. Exact monitors

Lemma 1 (Certified coverage; Ore [23] on nested graphs)**.** *The maximum number of tasks coverable by a feasible assignment is $M-d$, where*

$$d \;=\; \max_{1\le k\le M} \big(k - n_k\big)^+. \tag{3}$$

It is computable in $O(N\log N + M\log M)$, and a maximum feasible assignment is produced by processing tasks in ascending α and giving each one any *still-free eligible aircraft.*

Lemma 2 (Coverage margin; b-extended Hall [24])**.** *Full coverage survives the collapse ($\rho \to \infty$) of* any *b aircraft iff $n_k \ge k + b$ for all k. The largest such b is the* coverage margin *$\mu = \min_k(n_k - k)$, and full coverage currently fails iff $\mu < 0$.*

The margin is a procurement quantity. It counts the simultaneous certificate failures that the fleet absorbs with no loss of coverage. The next two results give the exact worst case when that budget is exceeded.

Proposition 1 (Optimal collapse adversary)**.** *If an adversary may collapse any b certificates, the maximum certified coverage afterwards is $M - d_b$ with $d_b = \max_k \big(k - (n_k - b)^+\big)^+$. The minimum is attained by collapsing the b aircraft with the smallest ρ.*

Proposition 2 (Graded degradation, exact worst case)**.** *Let any b aircraft have their protection level increased by $s \ge 0$, for example by s/L_ρ seconds of dead reckoning. Let $n_k^{(s)} = |\{i : \rho_i + s \le \alpha_{(k)}\}|$. Then the maximum certified coverage afterwards is at least*

$$M - \max_k \big(k - \max\{n_k - b,\, n_k^{(s)}\}\big)^+, \tag{4}$$

and an adversary choosing the b aircraft attains this value. As $s \to \infty$ it reduces to Proposition 1.

The generic bound "each failure costs at most one task", $M-d-b$, is valid but loose. Propositions 1 and 2 are exact, and on our fleets the gap is several tasks (Section V-F). Classical coverage control treats failed agents as binary [32]. Resilient multi-robot planning bounds submodular utility after α worst-case removals [33], [34]. Proposition 2 is exact for graded degradation, which is what jamming produces.

TABLE I
QUANTIFIERS ON A LEARNED CONSTRAINT. "EXOGENOUS" MEANS THE CONSTRAINT SET IS DRIVEN BY A PROCESS THE AGENT DOES NOT CONTROL; HERE, BY AN ADVERSARY.

Quantifier	Representative methods	What is guaranteed	Compromised policy (A4)
In expectation	CPO, PPO-/PID-Lagrangian, MACPO [12]–[14]	$\mathbb{E}[\sum_t \gamma^t c_t] \le d$, possibly after training	void
High probability, state-wise	state-wise CPO and variants [28], [29]	per-state constraint with probability $1-\delta$	void
Almost sure, own budget	Sauté RL, anytime-constrained RL [30], [31]	the agent's own cumulative cost never exceeds its budget	void
Every step, specified safe set	shields, safety layers, CBFs [15]–[18]	state stays in a set fixed by a specification or known dynamics; no utility guarantee	holds
Every epoch, certificate-driven set	**this work**	**(C) at every epoch for a set moved by an adversarial certificate; coverage maximal**	**holds**

B. The certified projection layer: a reference monitor

The projection sits in the environment, on the only path from a proposal to a flight command: the policy proposes, the environment executes Π of the proposal, and the policy is trained on what it proposed. Following Markgraf et al. [35], we call this the *safeguard-in-environment* (SE) placement, as opposed to placing the safeguard inside the policy.

Theorem 1 (Certified projection)**.** *Let Π be Algorithm 1 with $\eta = h = \ell_{\text{acq}} = \ell_{\text{keep}} = 0$. For every policy π_θ, every parameter value θ and every certificate trajectory permitted by Assumption 1, the executed assignment $\sigma_{t_k} = \Pi(a_{t_k}; s_{t_k})$ has the following properties:*

(i) it satisfies (C) at every epoch of every episode, in training and in deployment;

(ii) it assigns exactly $M - d(t_k)$ tasks, the maximum. Among maximum assignments it maximises assigned value, where $\min \Delta v$ is the smallest gap between distinct task values; among those it maximises agreement with the proposal a_{t_k};

(iii) if, in addition, the threat process is Markov in a state s that includes σ^-, the process seen by the policy is a Markov decision process (MDP) with kernel $P_\Pi(s' \mid s,a) = P(s' \mid s, \Pi(a;s))$. Its observations are partial when the policy is decentralised. The likelihood-ratio policy gradient of the projected objective is therefore unbiased, and PPO incurs no bias beyond its usual clipping and critic bias. Differentiating through Π instead gives zero gradient almost everywhere, because Π is piecewise constant in a.

For the coverage-only objective, the greedy nested variant of Lemma 1 achieves (i) and the cardinality part of (ii) in $O(N \log N + M \log M)$.

Two remarks place Theorem 1. First, part (i) alone is easy: the constraint is separable across aircraft, so per-agent action masking [36], [37] also never violates it. Theorem 1(i) is then the familiar shield guarantee, transposed to a certificate-driven set. What masking cannot provide is (ii). Agents that each pick a feasible task still collide, leaving coverage below $M - d$; Section V-C measures the difference. Second, the placement in (iii) follows the safeguard-in-environment analysis of [35]. That analysis shows that action aliasing, where many proposals map to one executed action, harms policy gradients more when the safeguard sits in the policy. Our setting has a further adversarial component: under Assumption 1 alone, the projected process is a game against the jammer rather than an MDP, and (i)–(ii) still hold while (iii) does not. We add a small penalty λ_{dev} on the fraction of proposals the projection overrides, so the policy learns to propose assignments the projection keeps.

C. Dynamics between and across epochs

Constraint (C) binds at decision epochs. Between epochs the certificate keeps moving, and across epochs an aircraft whose ρ_i hovers near α_j could acquire and drop a task repeatedly. We call a holder *exposed* on $[t_k, t_{k+1})$ if (C) held at t_k and fails before t_{k+1}. A *forced release* occurs when a holder becomes ineligible for its own task. Let $\text{TV}_i^+[0,\tau]$ be the positive variation of ρ_i on $[0,\tau]$: the sum of all its increases, including jumps.

Theorem 2 (Hysteresis)**.** *With band (η, h) in Algorithm 1 and $\ell_{\text{keep}} = 0$, the number of forced releases of aircraft i on $[0,\tau]$ is at most $1 + \text{TV}_i^+[0,\tau]/(\eta\,\alpha_{(1)} + h)$. If $\ell_{\text{keep}} \le \ell_{\text{acq}} + h$, the number of assigned feasible tasks is at least $M - d(\tilde{\rho})$, where $\tilde{\rho}_i = (\rho_i + h + \ell_{\text{acq}})/(1-\eta)$.*

Theorem 3 (Predictive feasibility and the allocator rate)**.** *Suppose that between upward jumps every ρ_i grows at rate at most L_ρ. If $\ell_{\text{keep}} \ge L_\rho \Delta t$ and $\ell_{\text{acq}} \ge L_\rho H$ with $H \ge \Delta t$, then:*

(i) a holder can be exposed on $[t_k, t_{k+1})$ only if its certificate jumps upward in that interval, so the number of exposed holder-epochs is at most the number of upward jumps;

(ii) a newly acquired task remains feasible for at least H unless a jump occurs.

An aircraft whose protection level is ρ can hold a task with alert limit α under this rule only if $\Delta t \le (\alpha - \rho)/L_\rho$.

With the measured constants, an aircraft at the benign ceiling $\bar{\rho}_0 = 0.94\,\text{m}$ can hold a 2 m corridor only if $\Delta t \le (2 - 0.94)/5.30 = 0.200\,\text{s}$. This is the allocator rate of 5.01 Hz quoted in the abstract: the rate at which *every* benign aircraft remains usable for the most demanding task. Aircraft with a lower benign level tolerate slower rates. Section V-H tests the rule at 2 Hz and at 5.3 Hz. Jumps cannot be predicted from a rate bound: a stale correction arrives all at once.

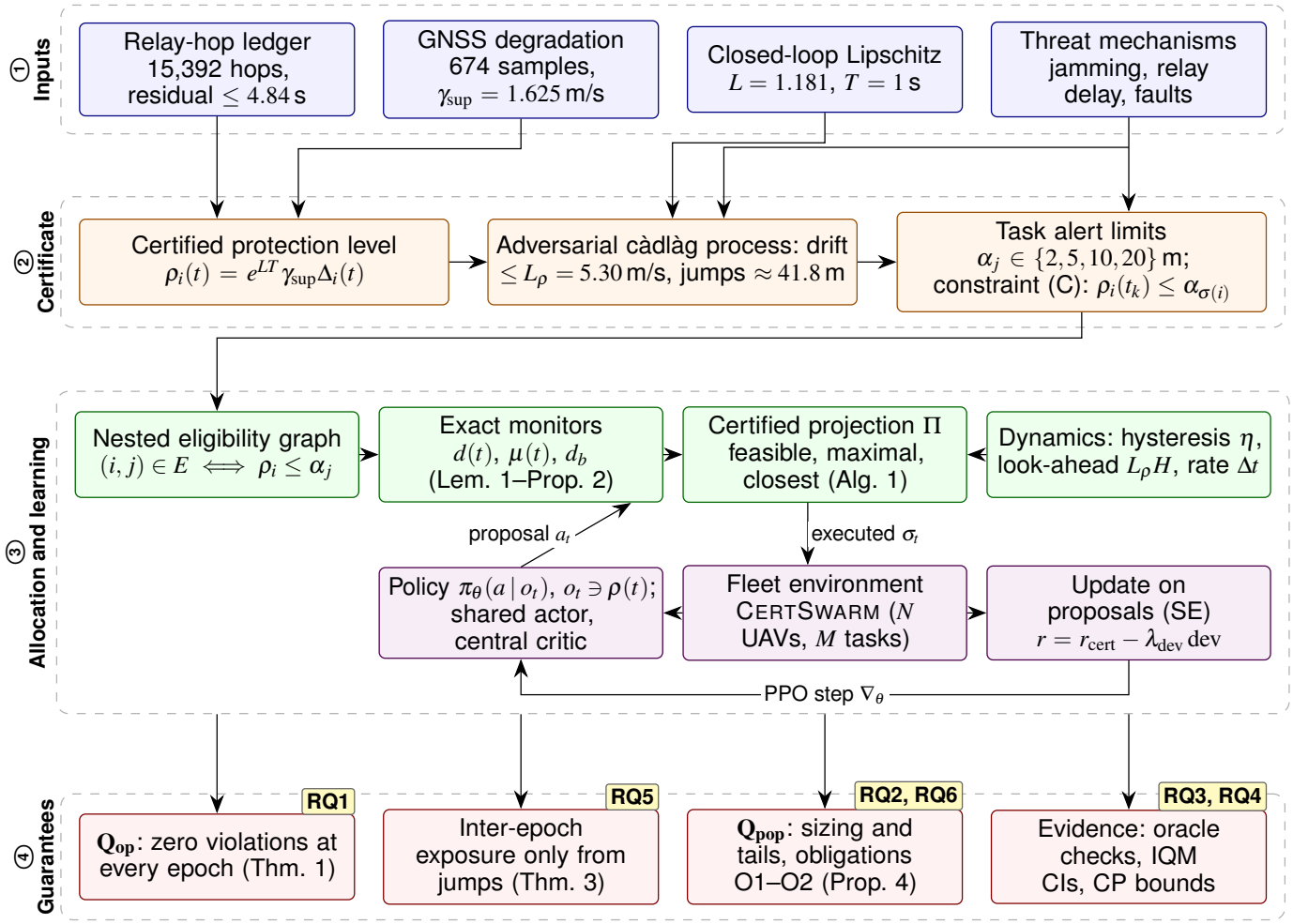


Fig. 1. Methodological framework, read top to bottom. ① Measured quantities from the parent artifact determine ② each aircraft's certified protection level $\rho_i(t)$, a bounded càdlàg process that drifts under jamming and jumps under relay delay; a task may be held only while $\rho_i(t)$ does not exceed its alert limit α_j. ③ Because the constraint is a scalar threshold on each side, the eligibility graph is nested. Closed forms monitor the coverage the fleet can certify, and a projection inside the environment maps every proposal of the learning policy to a feasible, coverage-maximal assignment, so training never executes an infeasible one. ④ The outputs are a per-operation guarantee and a population statement whose assumptions are listed as evidence obligations. Tags give the research question that each output answers.

Theorem 3 therefore isolates exactly what prediction can and cannot remove.

Proposition 3 (Misreported certificates)**.** *If f aircraft report a protection level other than their true one, the reported deficiency differs from the true one by at most f. Moreover, every violation of (C) under Π is committed by a misreporting aircraft.*

The deficiency depends on the certificates only through the count profile (n_k), which needs one counter per alert-limit class. Authenticated reports (Assumption 1(b)) reduce f to the number of compromised signing keys, but a Byzantine aircraft can still lie about its own certificate, equivocate, and drop the reports it relays. Without a trusted aggregator, the fleet must also agree on (n_k). We use a two-phase protocol: signed reports are flooded over the mesh, each honest aircraft forms its own count profile, and the profiles are reconciled by W-MSR resilient consensus [38], [39] with parameter $F = f$. W-MSR keeps every honest value inside the hull of honest initial values when the communication graph is $(2f+1)$-robust [38], so the agreed deficiency inherits the bound of Proposition 3. Section V-J measures what happens on meshes that are not robust enough.

D. From one operation to a population

Theorem 1 is a $\mathbf{Q}_{\text{op}}$ statement: it holds surely for one operation, given Assumption 1. Fleet sizing and per-flight-hour objectives need $\mathbf{Q}_{\text{pop}}$ statements, such as the probability that a fleet of size N covers every task at a random instant of a random operation. The companion systematisation shows that no chain of proofs alone reaches such an objective: every path crosses a statistical obligation [6]. The lift below names its own obligations.

Proposition 4 (Population lift)**.** *Group the alert limits into classes $\lambda_1 < \cdots < \lambda_c$ with m_ℓ tasks at or below λ_ℓ. Assume two obligations:*

Algorithm 1 Certified projection Π (executed at every decision epoch)

Require: proposal $a \in (\mathscr{T} \cup \{\varnothing\})^N$, certificates ρ, alert limits α, values v, previous assignment σ^-
Require: band $\eta \in [0,1)$, $h \geq 0$; look-aheads $\ell_{\text{acq}}, \ell_{\text{keep}} \geq 0$
$E_{ij} \leftarrow \left[\rho_i + \ell_{\text{acq}} \leq (1-\eta)\alpha_j - h\right]$ for $j \neq \sigma^-(i)$; $E_{i\sigma^-(i)} \leftarrow \left[\rho_i + \ell_{\text{keep}} \leq \alpha_{\sigma^-(i)}\right]$ ▷ acquire / keep
$W_{ij} \leftarrow B + c_v v_j + \mathbb{1}[a_i = j]$ on E ▷ $c_v = (N+1)/\min \Delta v$, $B > M(c_v \max v + 1)$
$\sigma \leftarrow$ maximum-weight matching of (E, W) ▷ Hungarian; greedy nested variant: Lemma 1
return σ ▷ executed; the policy is trained on a (SE placement)

O1 *at a random instant, and conditionally on the task set, the aircraft's classes $\ell_i = \min\{\ell : \rho_i \leq \lambda_\ell\}$ (with $\ell_i = c+1$, "none", if $\rho_i > \lambda_c$) are independent across aircraft;*

O2 *the classes are identically distributed with law q, with q estimated from operations representative of deployment.*

Then

$$\mathbb{P}\big(d = 0\big) = \mathbb{P}\Big(\textstyle\sum_{\ell' \leq \ell} C_{\ell'} \geq m_\ell \quad \forall \ell\Big),$$

where $C \sim \text{Multinomial}(N, q)$ over the $c+1$ classes; this is computable exactly by dynamic programming in $O(cN^2)$. The smallest fleet $N^\star(\delta)$ with $\mathbb{P}(d=0) \geq 1-\delta$ is the fleet-sizing answer, and the deficiency tail $\mathbb{P}(d \geq k)$ follows the same way.

O1 is precisely what a common-cause failure breaks. One jammer degrades every aircraft in its footprint at once. Section V-I shows that the independent model then *underestimates* multi-task loss, the operationally dangerous direction. Whenever zero violations are observed over n agent-epochs, we report the exact one-sided 95% Clopper–Pearson (CP) upper bound on the per-epoch violation rate [40]. It is approximately $3/n$, the "rule of three" [41] that also underlies the test-hour ladders of the Specific Operations Risk Assessment (SORA) [6], [42].

Remark 1 (Where the closed forms stop). Nestedness needs a scalar certificate against a scalar alert limit. With vector certificates (horizontal and vertical protection levels, link quality), eligibility becomes a dominance order. Maximum matching remains polynomial [43], but Equation (3) and the linear-time monitors are lost. Robust assignment under vertex failures is then hard to approximate in general [44]. The scalar case is therefore the largest one in which every monitor is a closed form.

IV. Evaluation Methodology

a) Certificate process.: We generate $\rho_i(t)$ by (1) from three independent mechanisms per aircraft:

- dead reckoning under jamming or sensor faults: staleness grows at 1 s/s, Poisson onsets, exponential durations, and a 2 s re-anchoring delay;
- undetected relay-delay episodes over $H = 2$ hops: staleness jumps by the sum of two residuals drawn from the 1,346 measured undetected residuals;
- a benign floor drawn from the 10,621 measured benign residuals.

Spatial jammers (radius 200 m, random walk) add common-cause dead reckoning to every aircraft in their footprint. Rates are stated modelling assumptions and are swept; every magnitude is measured. Appendix A lists each constant with its provenance.

b) Environment.: CertSwarm is a vectorised environment of B fleets with $N = 12$ aircraft and $M = 8$ tasks on a 1×1 km arena. Aircraft fly at 15 m/s, and a task is on site within 30 m. Two spatial jammers roam the arena. Decision epochs are $\Delta t = 0.5$ s, and missions last 150 s. Alert limits are drawn from $\{2, 5, 10, 20\}$ m with values $\{4, 3, 2, 1\}$.

Each aircraft observes:

- its own position, ρ_i, the change in ρ_i, and whether it holds a task;
- for each task, in egocentric order of distance: relative position, alert limit, value, eligibility, and whether it or another aircraft holds the task;
- the fleet's count profile and the jammer offsets.

The jamming-blind variant zeroes every certificate-derived feature.

c) Learning.: We use proximal policy optimisation (PPO) [45] in the multi-agent PPO (MAPPO) style [46]. A parameter-shared actor outputs, per aircraft, a categorical over {keep, idle, k-th nearest task}. A centralised critic sees the aircraft's observation and the fleet mean. Training uses 16 parallel fleets, rollouts of 150 epochs and 200 updates (480k fleet-epochs, 5.8×10^6 agent-epochs per run), with 10 training seeds per method. Appendix A lists all hyperparameters.

d) Methods.:

- *Ours*: Algorithm 1 in the environment, reward r_{cert}, and deviation penalty $\lambda_{\text{dev}} = 0.05$. The ablation sets $\lambda_{\text{dev}} = 0$.
- *Masking*: invalid-action masking per aircraft [36], with conflicts resolved by the first claimant.
- *PPO-Lagrangian* [13], [47]: the certificate is observed but not enforced. The reward is nominal coverage minus λ times the violation rate; λ is updated by dual ascent to a limit of 0.5% violating agent-epochs.
- *Jamming-blind PPO*: no certificate features, nominal reward.
- Every learned policy without the layer is also evaluated *with the projection added at deployment*, separating what the policy learns from what the mechanism enforces.
- Non-learned references:
  - keep-or-nearest proposals with the projection;
  - the tightest-fit greedy allocator re-solved every epoch;
  - a fixed design margin that admits task j iff $\rho_0 + L_\rho \cdot 1\,\text{s} \leq \alpha_j$, i.e. one second of dead reckoning budgeted at design time;
  - keep-or-nearest with no certificate.

e) Stronger in-expectation and allocation baselines.: At the headline size we add, in the same environment and with the same budget of environment steps:

- *MAPPO-Lagrangian* with a separate cost critic and cost advantage, as in SafePO [14], [48], at cost limits 0.005 and 0;
- *PID-Lagrangian PPO* [13], parameter-shared;
- *QMIX* [49] with a monotonic hypernetwork mixer and double-Q targets, with the certificate observed, and with the per-aircraft eligibility mask in acting and in its targets;
- the *consensus-based bundle algorithm* (CBBA) [50] with bundle length one and feasibility enforced inside the bundle. On a connected network its consensus converges to the sequential greedy auction, which we compute directly; the bid is task value minus a distance cost.

Masking with conflict resolution is the factored shield of [16], trained here with PPO rather than MADDPG. We did not run MACPO; its second-order update was not reimplemented, and we do not report it.

f) Fleet size.: We scale to $N \in \{24, 48, 96\}$ with $M = \lceil 2N/3 \rceil$, the arena side scaled by $\sqrt{N/12}$ and spatial jammers in proportion to $N/12$. To hold the agent-samples per update fixed, the number of parallel fleets falls as $192/N$. Up to $N = 48$ the actor is the same multilayer perceptron; at $N = 96$ it is a permutation-invariant pointer network that attends over the 16 nearest tasks, so its input does not grow with M. Budget: 200 updates and 5 seeds per method for $N \leq 48$, 150 updates and 3 seeds at $N = 96$. Learned methods are the four core ones; the non-learned references are re-run at every size.

g) Byzantine aircraft and agreement (RQ7).: For $N = 12$, $M = 8$ and $f \in \{0, 1, 2\}$ liars, we sample meshes with the platform's mesh generator (a ring plus chords, at three densities) and the complete graph, and compute each graph's robustness exactly by enumerating all 3^{12} pairs of disjoint node sets. Liars inflate their eligibility (report 0), deflate it (report ∞) or equivocate, and they either forward or drop the reports they relay. We run the two-phase protocol of Section III with $F = f$ and measure agreement, rounds, the error in the agreed deficiency and violations by honest aircraft. We use 200 instances per cell.

h) Metrics and statistics.: We report:

- certified coverage r_{cert};
- the *coverage ratio*, feasible holders over the maximum $M - d(t)$ from Lemma 1, which separates "safe because idle" from "safe and maximal";
- violations per agent-epoch, counted per step and never averaged over episodes, in training and in evaluation;
- inter-epoch exposure and reassignments per agent-epoch.

Each learned method is evaluated greedily on 3 held-out seeds $\times$ 16 fleets per training seed. We report the interquartile mean (IQM) with 95% stratified-bootstrap confidence intervals (CIs) [51], and CP bounds whenever zero violations are observed.

V. Evaluation Results

A. Correctness of the closed forms

Over 10,000 random nested instances (up to 10 aircraft and 10 tasks, with ties, continuous and heavy-tailed certificates), we compared every closed form with a brute-force oracle: Hopcroft–Karp matching [43], enumeration of all b-subsets, and enumeration of task subsets for value. The checks covered:

- Lemma 1, the greedy and lexicographic projections, and value optimality;
- Lemma 2 and Proposition 1;
- the coverage bound of Theorem 2.

The 9 checks total 83,872 comparisons with 0 disagreements. A further 4,000 instances per misreporting level confirm Proposition 3: no honest aircraft ever violated (C). This is a check on the mathematics and the implementation, not evidence about flight.

B. RQ1: violations during learning

Figure 2 tracks certified coverage and cumulative violations over training.

- *Projection in the environment.* It executed *zero* infeasible assignments over 5.8×10^7 agent-epochs, pooled across 10 seeds, from the first update onwards. Per-agent masking also reached zero, as separability predicts. For both methods zero holds by construction, so the CP bound in Table II only quantifies the exposure observed. That bound assumes independent agent-epochs, which these are not.
- *PPO-Lagrangian.* Although it observes the certificate, it executed 8,193,222 infeasible assignments while learning, 14.22% of agent-epochs. Its multiplier then rose until the policy largely stopped serving tasks. Final certified coverage is 0.117, and evaluation violations still average 2.24% of agent-epochs, above the 0.5% limit.
- *Jamming-blind PPO.* It executed 9,506,180 infeasible assignments (16.50%).
- *Stronger in-expectation learners* (Table III). MAPPO-Lagrangian with a cost critic executed 2,897,602 infeasible assignments while learning (5.03%), and 2,929,578 with a cost limit of zero. PID-Lagrangian executed 8,089,556 and then stopped serving tasks altogether. QMIX without masks executed 7,526,523. Only the two methods that remove infeasible actions by construction, our layer and masking (with PPO or with QMIX), learned without a violation.

In-expectation methods violate the constraint while they learn, and learning is when a fleet would be flight-tested. No setting of the Lagrangian's cost limit or dual step changed this (Appendix A).

C. RQ3: what the policy learns and what the mechanism enforces

Table II evaluates every method greedily on held-out missions. Four findings stand out.

The mechanism does most of the work. Adding the projection at deployment to *any* learned policy removed every violation and lifted the coverage ratio to at least 0.99:

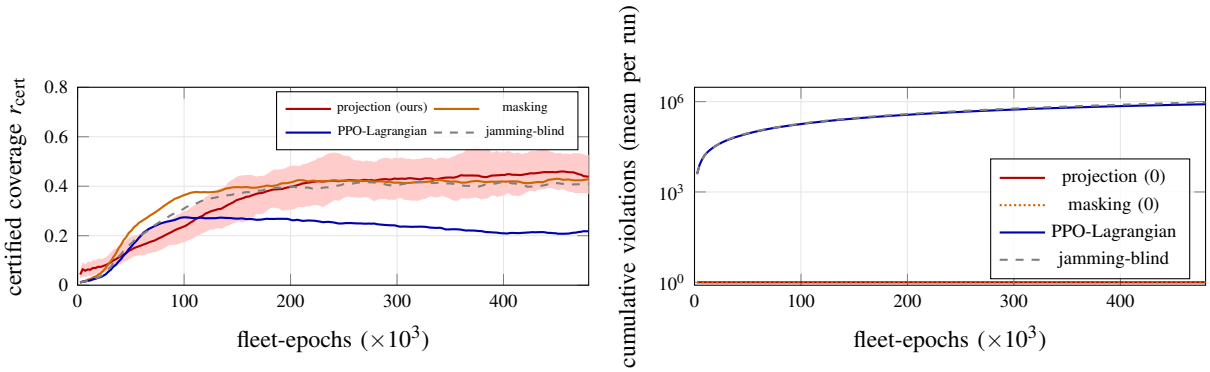


Fig. 2. RQ1. Left: certified coverage during training (stochastic policy; mean over 10 seeds, 10-update moving average; band: min–max across seeds for ours). Right: cumulative executed violations on a log axis. Zero is drawn at 1 for visibility. The projection and masking curves lie on the floor throughout.

- jamming-blind: certified coverage $0.435 \to 0.491$;
- masking: $0.464 \to 0.498$;
- PPO-Lagrangian: $0.117 \to 0.431$.

Without it, the jamming-blind policy and the no-certificate heuristic reach seemingly competitive certified coverage (0.435 and 0.459). They do so only by holding tasks without certification on 12.30% and 13.97% of agent-epochs. The jamming-blind policy in fact converges to the keep-or-nearest rule, with no reassignments.

Masking is safe but not maximal. Its coverage ratio of 0.62 is the gap that Theorem 1(ii) closes. Agents that each pick an eligible task still collide.

Learning adds little on top of the guarantee, and where the layer sits during training does not matter.

- The policy trained with the projection reaches 0.491 [0.477, 0.509] at a coverage ratio of 1.00, over 10 training seeds.
- The keep-or-nearest heuristic with the same projection reaches 0.435; the intervals only just separate.
- Policies trained with masking, or blind, and projected at deployment do as well (0.498, 0.491), and the deviation penalty makes no measurable difference (0.503 without it).

What training *with* the projection buys is the zero-violation learning of RQ1, not a better final policy.

Static and design-time allocators fail in opposite ways. The tightest-fit greedy re-solves from scratch every epoch, keeps its aircraft in transit (8.6% reassignments) and certifies only 0.169. A fixed design margin certifies 0.163 and still violates on 8.03% of agent-epochs, because a margin fixed at design time cannot track a certificate that moves.

D. RQ3 (continued): stronger baselines and a classical auction

Table III adds the stronger learners and a non-learned consensus auction at the headline size. Three findings matter for the security argument.

Better in-expectation learners still violate, and the monitor still repairs them. MAPPO-Lagrangian with a cost critic is the strongest unconstrained learner: it reaches 0.445 certified coverage and, once trained, violates on only 0.027% of agent-epochs. Its guarantee is nonetheless in expectation: it paid for that policy with 2,897,602 violations during learning, and lowering its cost limit to zero did not remove them. With the projection at deployment, every learned policy in the table executes no violation and reaches a coverage ratio of at least 0.98.

A classical auction under the same constraint beats every learned allocator. CBBA with feasibility enforced inside the bundle certifies 0.539 [0.525, 0.561] of the task value, against 0.491 [0.477, 0.509] for the best learned policy, with no violation and a coverage ratio of 1.00. The comparison favours the auction in one respect: its consensus phase gives every aircraft the fleet's full state at every epoch, while the learned policies act on local observations in one shot. In our setting, that information is worth more than learning.

The result supports the paper's claim, not a learning claim. Our guarantees do not depend on which allocator proposes: they hold for the auction, for the heuristic, for every learned policy and for an adversarially chosen one (A4). The learning study shows that the monitor can sit in the training loop at no cost to learning; it does not show that learning is the best allocator, and we do not claim it.

E. RQ3 (continued): fleet size

Table IV and Figure 3 repeat the core methods at $N \in \{12, 24, 48, 96\}$.

- *The guarantees scale.* Our layer and masking executed no infeasible assignment during training at any size, and our coverage ratio was at least 1.00 throughout. PPO-Lagrangian executed millions of violations at every size and, from $N = 24$ on, converged to idling (0.000 certified coverage).
- *Learning does not scale with our budget, and the projection's objective is part of the reason.* Certified coverage of the learned policy fell from 0.491 at $N = 12$ to 0.386 at $N = 48$ and 0.009 at $N = 96$, where aircraft were reassigned on 36.2% of agent-epochs and rarely reached a site. Two causes compound. First, a confound of ours: the actor's keep prior was a fixed logit, so the initial probability of keeping a task fell as the action set grew. Scaling the prior to hold the $N = 12$ value changed coverage only to 0.407 at $N = 48$ and 0.014 at $N = 96$. Second, Algorithm 1 maximises cardinality, then value, and only then agreement with the proposal, and

TABLE II

RQ3. HELD-OUT EVALUATION ($N = 12$, $M = 8$; 10 TRAINING SEEDS $\times$ 3 EVALUATION SEEDS $\times$ 16 FLEETS FOR LEARNED METHODS; 3×16 FOR THE REFERENCES BELOW THE LINE). VALUES ARE IQMS WITH 95% STRATIFIED-BOOTSTRAP CIS. COV. RATIO: FEASIBLE HOLDERS DIVIDED BY $(M - d)$. VIOLATIONS ARE IN % OF AGENT-EPOCHS (IQM OVER RUNS). "$0\,(\leq x)$": NONE OBSERVED, WITH THE 95% CP UPPER BOUND ON THE RATE, ASSUMING INDEPENDENT AGENT-EPOCHS. TRAIN VIOL.: EXECUTED INFEASIBLE ASSIGNMENTS OVER ALL OF TRAINING.

Method	r_{cert}	cov. ratio	eval viol. [%]	train viol.	exposure [%]	reassign. [%]
Projection (SE) + PPO (ours)	0.491 [0.477, 0.509]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	$0\ (\leq 5.2 \times 10^{-8})$	1.66	4.9
ablation: no deviation penalty	0.503 [0.493, 0.516]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	$0\ (\leq 5.2 \times 10^{-8})$	1.67	4.9
Per-agent action masking + PPO	0.464 [0.459, 0.471]	0.62	$0\ (\leq 1.7 \times 10^{-6})$	$0\ (\leq 5.2 \times 10^{-8})$	0.53	0.7
+ projection at deployment	0.498 [0.492, 0.505]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	–	1.69	4.8
PPO-Lagrangian (certificate observed)	0.117 [0.105, 0.131]	0.13	1.17	8,193,222	0.09	0.0
+ projection at deployment	0.431 [0.418, 0.444]	0.99	$0\ (\leq 1.7 \times 10^{-6})$	–	1.51	4.7
Jamming-blind PPO	0.435 [0.425, 0.447]	0.49	12.95	9,506,180	0.36	0.0
+ projection at deployment	0.491 [0.480, 0.503]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	–	1.68	5.0
Keep-or-nearest + projection	0.435 [0.395, 0.477]	1.00	$0\ (\leq 1.7 \times 10^{-5})$	–	1.45	2.3
Tightest-fit greedy, re-solved each epoch	0.169 [0.142, 0.216]	0.98	$0\ (\leq 1.7 \times 10^{-5})$	–	1.99	8.6
Fixed design margin (1 s dead reckoning)	0.163 [0.155, 0.170]	0.31	8.03	–	0.20	0.0
Keep-or-nearest, no certificate	0.459 [0.433, 0.484]	0.51	13.97	–	0.38	0.0

TABLE III

RQ3. STRONGER BASELINES AT $N = 12$ (10 TRAINING SEEDS $\times$ 3 EVALUATION SEEDS $\times$ 16 FLEETS; CBBA IS DETERMINISTIC, 3×16). COLUMNS AS IN TABLE II; EVAL. VIOLATIONS IN % OF AGENT-EPOCHS.

Method	r_{cert}	cov. ratio	eval viol. [%]	train viol.
Projection (SE) + PPO (ours)	0.491 [0.477, 0.509]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	$0\ (\leq 5.2 \times 10^{-8})$
Keep-or-nearest + projection	0.435 [0.395, 0.477]	1.00	$0\ (\leq 1.7 \times 10^{-5})$	–
MAPPO-Lagrangian, cost critic, limit 0.005	0.445 [0.438, 0.451]	0.57	0.027	2,897,602
+ projection at deployment	0.463 [0.455, 0.473]	0.99	$0\ (\leq 1.7 \times 10^{-6})$	–
MAPPO-Lagrangian, cost critic, limit 0	0.442 [0.436, 0.451]	0.57	0.091	2,929,578
+ projection at deployment	0.466 [0.455, 0.477]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	–
PID-Lagrangian PPO	0.000 [0.000, 0.000]	0.00	0.040	8,089,556
+ projection at deployment	0.347 [0.336, 0.360]	0.98	$0\ (\leq 1.7 \times 10^{-6})$	–
QMIX (certificate observed)	0.422 [0.410, 0.436]	0.60	8.473	7,526,523
+ projection at deployment	0.460 [0.446, 0.477]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	–
QMIX with eligibility masks	0.451 [0.444, 0.459]	0.68	$0\ (\leq 1.7 \times 10^{-6})$	$0\ (\leq 5.2 \times 10^{-8})$
+ projection at deployment	0.455 [0.437, 0.473]	1.00	$0\ (\leq 1.7 \times 10^{-6})$	–
CBBA, feasibility in the bundle	0.539 [0.525, 0.561]	1.00	$0\ (\leq 1.7 \times 10^{-5})$	–

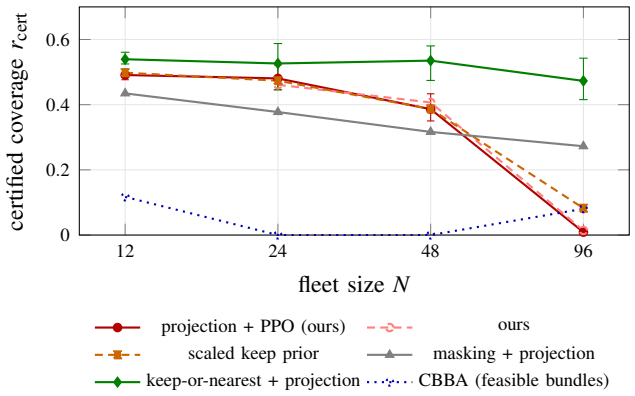


Fig. 3. RQ3, fleet size. Every method shown except PPO-Lagrangian executes no infeasible assignment. At a fixed training budget the learned allocators lose coverage as the fleet grows, collapsing at $N = 96$ through reassignment churn; the auction does not.

none of the three terms sees travel distance. With a weak proposal on a large fleet it moves aircraft between equally valued tasks, which keeps every guarantee (zero violations, coverage ratio 1.00) but not the aircraft on site. A keep term ($\varepsilon_{\text{prev}} > 0$ in Algorithm 1) or a distance-aware value is the obvious remedy; we did not evaluate it.

- *The auction does scale.* CBBA certified 0.539, 0.526, 0.535 and 0.473 of the task value at the four sizes, ahead of keep-or-nearest plus projection (0.272 at $N = 96$).

Our budget per size was 200 updates (150 at $N = 96$) on 4 CPU cores, below the runbook's 400–600; larger budgets may narrow the gap to the auction, and the artifact runs them unchanged.

F. RQ2: coverage as certificates degrade

Figure 4 shows fleets caught at a random instant of an operation, with heterogeneous certificates drawn from the stationary certificate process. Under binary collapse (left), the exact worst case of Proposition 1 lies well above the generic bound. At $\phi = 0.3$ ($b = 4$ of 12 aircraft), the fleet still certifies 5.55 tasks against the worst adversary and 6.37 under random collapse. The generic bound promises only 3.89. Under graded degradation of 1 s of dead reckoning (right), Proposition 2 coincided with the brute-force worst case in 100% of instances, with 0 violations of the bound.

G. RQ4: scalability

Figure 5 reports wall-clock latency on an otherwise idle 4-vCPU cloud server (AMD EPYC-Genoa Processor). The closed-form monitor (deficiency and margin) takes 0.058 ms at $N = 1000$ and 11.0 ms at $N = 10^5$. The greedy nested projection takes 0.17 ms at $N = 1000$ and 26 ms at $N = 10^5$. The exact lexicographic projection (Hungarian) is the costly component: 0.018 ms for our 12-aircraft fleets, but 11.4 ms at $N = 1000$ and 30 ms at $N = 2000$. Generic Hopcroft–Karp matching takes 671 ms at $N = 10^4$. Against the 0.200 s epoch of Theorem 3, the exact layer is real-time up to the largest size we timed ($N = 2000$), and the greedy layer beyond 10^5. Beyond the exact layer's range, the value and agreement parts of Theorem 1(ii) must be dropped or approximated; the cardinality guarantee is kept.

TABLE IV

RQ3, FLEET SIZE. CERTIFIED COVERAGE (IQM, 95% STRATIFIED-BOOTSTRAP CI) OF THE LEARNED POLICY WITH THE PROJECTION IN THE ENVIRONMENT (KEEP PRIOR FIXED AT +3, AND SCALED TO KEEP THE $N = 12$ INITIAL KEEP PROBABILITY), MASKING WITH THE PROJECTION AT DEPLOYMENT, AND TWO NON-LEARNED ALLOCATORS; $M = \lceil 2N/3 \rceil$, ARENA AND JAMMERS SCALED WITH N. COV. RATIO IS THAT OF THE SCALED-PRIOR POLICY.

N	seeds	ours, prior +3	ours, scaled prior	cov. ratio	masking + proj.	keep-or-nearest + proj.	CBBA	Lagrangian train viol.
12	10	0.491 [0.477, 0.509]	0.491 [0.477, 0.509]	1.00	0.498 [0.492, 0.505]	0.435 [0.395, 0.477]	0.539 [0.525, 0.561]	8,193,222
24	5	0.481 [0.446, 0.522]	0.461 [0.433, 0.507]	1.00	0.474 [0.437, 0.516]	0.377 [0.343, 0.439]	0.526 [0.446, 0.588]	4,583,986
48	5	0.386 [0.350, 0.433]	0.407 [0.374, 0.453]	1.00	0.387 [0.353, 0.439]	0.316 [0.246, 0.401]	0.535 [0.474, 0.580]	4,685,248
96	3	0.009 [0.008, 0.011]	0.014 [0.010, 0.024]	1.00	0.082 [0.077, 0.088]	0.272 [0.197, 0.316]	0.473 [0.416, 0.543]	1,976,207

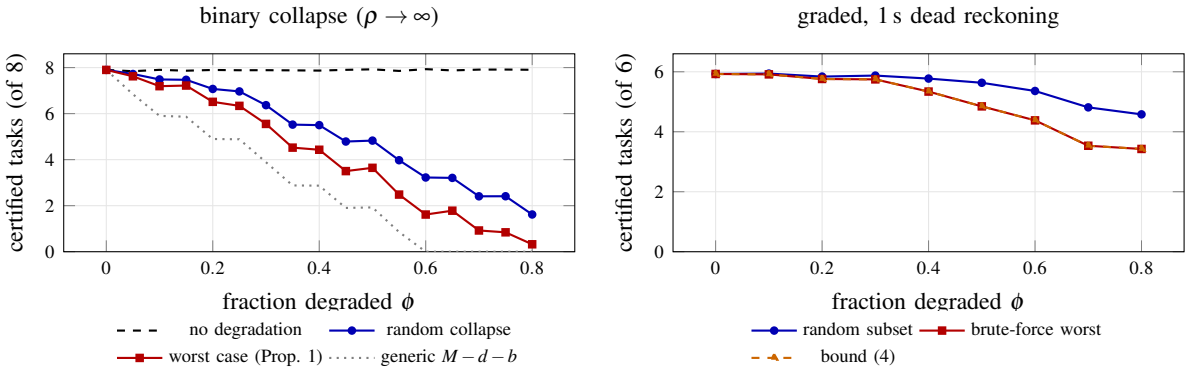


Fig. 4. RQ2. Coverage as a fraction ϕ of the fleet degrades. Left: $N = 12$, $M = 8$, 400 fleets per point. Right: $N = 10$, $M = 6$, 150 fleets per point with exhaustive worst case; the bound of Proposition 2 lies on the worst case throughout.

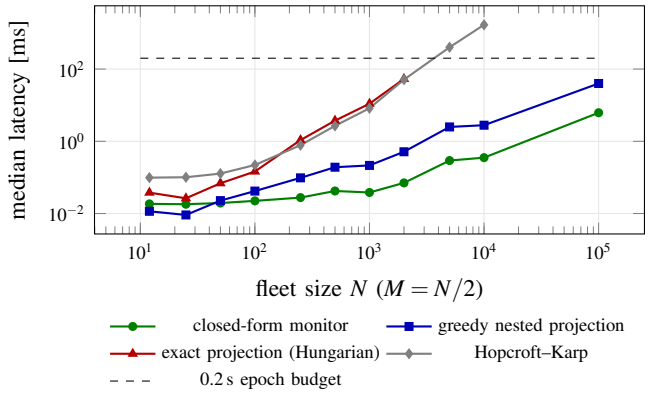


Fig. 5. RQ4. Latency of the monitor and the projection layer against fleet size.

TABLE V

RQ5. PROJECTION RULES BETWEEN DECISION EPOCHS (16 FLEETS × 4 SEEDS, 150 S MISSIONS). EXPOSURE: HOLDERS FEASIBLE AT AN EPOCH BUT NOT BEFORE THE NEXT. "NO JUMP": EXPOSURES WITH NO UPWARD CERTIFICATE JUMP IN THE INTERVAL.

Rule	Δt	r_{cert}	cov. ratio	exposure [%]	exposures w/o jump
standard (C) only	0.5 s	0.393	0.983	1.51	2930
predictive release only	0.5 s	0.209	0.983	1.18	2182
predictive, $H = \Delta t$	0.5 s	0.263	0.834	0.19	0
predictive, $H = 1$ s	0.5 s	0.168	0.654	0.14	0
predictive, $H = 2$ s	0.5 s	0.083	0.377	0.10	0
standard (C) only	0.19 s	0.390	0.977	0.56	2842
predictive, $H = \Delta t$	0.19 s	0.371	0.983	0.09	0

H. RQ5: between decision epochs

Table V runs the projection with keep-or-nearest proposals, isolating the allocation layer. At $\Delta t = 0.5$ s, the standard rule leaves 1.51% of agent-epochs exposed, 2930 of them without any certificate jump. The predictive rule of Theorem 3 cuts exposure by 8.1× and leaves 0 exposures without a jump. In our simulator the drift between jumps equals L_ρ exactly and ρ is constant between epochs, so this zero is a sanity check of the implementation rather than evidence about flight. The rule also costs coverage, $0.393 \to 0.263$, because at 2 Hz the 2 m tasks become unholdable. Releasing early without predicting acquisitions ("release only") is worse: aircraft acquire, release and reacquire, and coverage drops to 0.209. At $\Delta t = 0.19$ s (5.3 Hz), just inside the rate of Theorem 3, the predictive rule costs little: coverage is 0.371 against 0.390. Exposure falls by 6.4×, and every remaining exposure is again caused by a jump. Hysteresis was much less useful here:

- the bound of Theorem 2 held for 100% of aircraft;
- a band of $\eta = 0.5$ reduced reassignments from 2.38% to 2.01% of agent-epochs, at a coverage cost of $0.393 \to 0.379$.

The reason is that the certificate moves in steps of $L_\rho \Delta t \approx 2.6$ m, which already exceed the bands of the demanding tasks. Chattering matters for smooth certificates; this one moves in steps.

I. RQ6: the population lift and its obligation

We run 64 fleets ($N = 12$, $M = 8$, fixed alert limits) for 200 s, discard the first 25 s, and estimate the class law q from the remaining 175 s. We then compare the deficiency distribution that Proposition 4 predicts from q with the simulated one

(Figure 6). Because q is estimated on the same runs, the marginals are matched by construction and only the dependence structure is tested.

- *Independent degradation, O1 true* (per-aircraft jamming onsets, no spatial jammers). The lift is accurate: $\mathbb{P}(d=0)$ is 0.934 predicted against 0.927 simulated, and every tail agrees within its CI.
- *Common-cause jamming, O1 false.* Aircraft loiter around three operating areas while three jammers of radius 260 m drift at 4 m/s. At a random instant, 58% of aircraft are certified for no task. The independent model is now wrong in both directions:
  - it predicts full coverage 0.074 of the time against 0.250 [0.175, 0.334], too pessimistic by 3.4×;
  - it predicts that six or more of the eight tasks are lost 6.5% of the time against 24.1% [15.5, 32.9], too optimistic by 3.7×.

Correlation widens the distribution: jammers that miss a cluster leave it whole, and jammers that hit it take it all. A fleet sized by the independent model, $N^\star = 15$ aircraft for 99% full coverage under independent degradation, is therefore not sized for the catastrophic losses that a common-cause jammer produces. O1 must be evidenced by spatial separation or by a common-cause analysis, not assumed. This is the obligation-carrying lift argued for in [6], made quantitative.

a) Measured traces (runbook R3).: We replaced the Poisson jamming and fault onsets by GNSS staleness replayed from 11 PX4 flight logs of a public UAV attack dataset [52] (3 benign, 1 jamming, 3 spoofing and 4 link-DoS flights), measured over each flight's airborne window as the time since the last GNSS fix the estimator fused. The relay channel keeps its measured magnitudes, because the hop ledger holds one packet per run and therefore no time series. Three observations follow.

- *Real jamming was brief.* The live jamming flight lost its fix once for more than 1 s (at most 2.0 s, 0.46% of airborne epochs). Staleness in the simulated flights never exceeded 0.30 s. The Poisson model of Section IV is therefore a stress test, not a description of these flights.
- *Fix age is not negligible.* On the live hardware the median staleness was 0.59–0.85 s, consistent with a 1 Hz receiver. Through L_ρ that alone is up to 4.5 m of certified error, enough to make the 2 m tasks unholdable for that receiver.
- *The independence obligation still fails.* Without spatial jammers the traces make aircraft independent by construction, and the lift is exact (0.998 predicted against 0.997 simulated). With common-cause jammers added, the independent model predicts losing six or more tasks 6.5% of the time against 20.8% simulated, 3.2× too low, as with synthetic onsets.

In the learning study on traces the ordering is unchanged: CBBA 0.546, masking with projection 0.517, our layer 0.505 [0.495, 0.513] with zero training violations, keep-or-nearest with projection 0.447, and PPO-Lagrangian idling after 2,649,906 training violations.

TABLE VI

RQ7. TWO-PHASE AGREEMENT WITH f BYZANTINE AIRCRAFT ($N=12$, 200 INSTANCES PER LIAR STRATEGY AND RELAY BEHAVIOUR). ROB.: MEDIAN ROBUSTNESS OF THE SAMPLED MESHES; AGREED, EQUIV.: W-MSR AGREEMENT WITH EQUIVOCATING LIARS; $\max|\Delta d|$: WORST ERROR IN THE AGREED DEFICIENCY OVER ALL LIAR STRATEGIES, WHEN LIARS FORWARD OR DROP RELAYED REPORTS.

mesh	f	rob.	$(2f{+}1)$-rob. [%]	agreed, equiv. [%]	max $\lvert\Delta d\rvert$ fwd	max $\lvert\Delta d\rvert$ drop	honest viol.
ring + chords (sparse)	1	1	0	66	1	1	0
	2	1	0	26	2	4	0
ring + chords (medium)	1	2	0	96	1	1	0
	2	2	0	51	2	2	0
ring + chords (dense)	1	3	55	100	1	1	0
	2	3	0	74	2	2	0
complete	1	6	100	100	1	1	0
	2	6	100	100	2	2	0

J. RQ7: Byzantine aircraft and decentralised agreement

Table VI runs the two-phase protocol with up to two liars.

- *Integrity holds unconditionally.* Across every mesh, liar strategy and relay behaviour, honest aircraft executed 0 violations; all 4,819 violations were committed by liars, as Proposition 3 states.
- *The monitor's error is bounded by f when reports get through.* When liars forward the reports they relay, the agreed deficiency never departs from the truth by more than f, on every mesh. Whenever W-MSR reached agreement, the error was within f.
- *Equivocation needs a robust mesh.* On the complete graph (robustness 6) W-MSR agreed in every instance within 6 rounds. The platform's ring-plus-chords meshes have median robustness 1, 2 and 3 at the three densities; with two equivocating liars they agreed in only 26%, 51% and 74% of instances. The failure is the one the theory predicts: these graphs are not $(2f+1)$-robust.
- *Dropping is outside Proposition 3.* When liars also drop the reports they relay, sparse meshes lose honest reports (24% of instances on the ring with $f=2$), and the deficiency error reaches 4, twice the bound. Integrity is unaffected, because the monitor only under-assigns, but availability is.

The design consequence is concrete: a fleet that must tolerate f compromised aircraft needs a mesh whose robustness is at least $2f+1$, and the closed-form monitor makes it cheap to check the counts the mesh agrees on.

a) Vector certificates (Remark 1).: We gave each aircraft horizontal and vertical protection levels, with the vertical-to-horizontal ratio lognormal around 1.5, and each task horizontal and vertical alert limits. The scalar surrogate $\max(\rho_h, \rho_v/c)$ against $\min(\alpha_h, \alpha_v/c)$ is sound: over 400 instances per cell it never admitted an infeasible pair, and its closed-form worst case always bounded the brute force. It is conservative. At $N=12$ and dispersion 0.5, only 74% of exact eligibility graphs were still nested, and the surrogate lost 0.65 of 8 tasks on average (some loss in 49% of instances). At $N=20$ and dispersion 1, 10% were nested and it lost 1.73 tasks on average, up to 5. The exact worst case after three collapses, by brute force, took 4.9 ms at $N=8$ and 106 ms at $N=20$, against at most 0.04 ms for the closed form. With perfectly correlated levels the loss was 0.0.

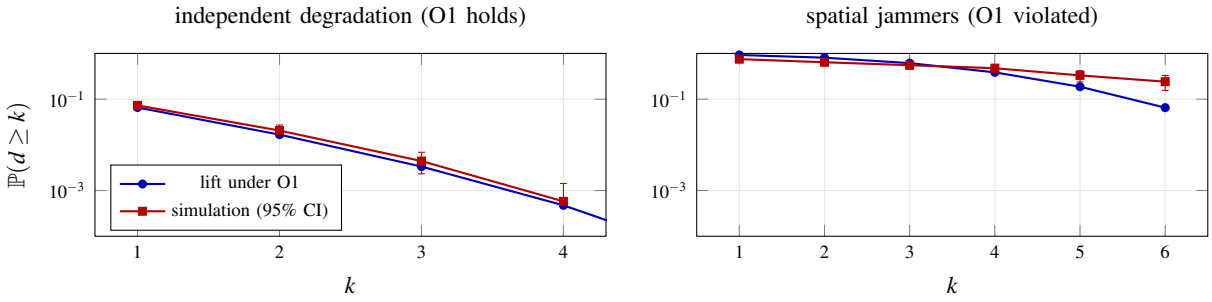


Fig. 6. RQ6. Deficiency tail predicted by the population lift (Proposition 4) against simulation, $N = 12$, $M = 8$; CIs bootstrap over 64 independent fleets. Missing points are zero. The lift fails exactly where its independence obligation fails.

VI. Related Work

a) Reference monitors and runtime assurance.: Our layer is a reference monitor in Anderson's sense [19]: always invoked, small enough to verify, and the only path from a proposal to an executed command. Runtime-assurance architectures, from Simplex [20] to the ASTM practice for aircraft with complex functions [21], switch to a verified fallback when a complex controller leaves a safe envelope. We do not switch: the monitor repairs every proposal to the nearest certified, coverage-maximal assignment, so the fallback is never idle unless the certificates leave nothing to cover.

b) Attacks on learned controllers and on navigation.: Neural policies can be steered by adversarial observations [7], by adversarial co-players [8] and by backdoors planted during training [9], [10]. Certified RL bounds the effect of a bounded perturbation on one policy [53], [54]; our guarantee needs no bound on the perturbation, because it holds for every policy parameter. GNSS spoofing and jamming [1]–[3] motivate the certificate we consume [4]. Byzantine agreement [55], approximate agreement [39] and W-MSR resilient consensus [38] give the decentralised route to the monitor's inputs; we show where the robustness condition of the last one fails on sparse UAV meshes.

c) Constrained and safe RL.: CMDPs [11] and their solvers are reviewed in [56], [57]:

- trust-region and primal-dual methods [12], [13], [58], [59], with zero duality gap [60] and near-optimal sample complexity [61];
- state-wise and high-probability formulations [28], [29], [62];
- almost-sure and anytime constraints [30], [31], [63];
- instantaneous hard constraints under unknown safety [64].

Anytime-constrained RL is the closest quantifier, and computing optimal policies there is NP-hard [31]. Its cost is generated by the agent's own trajectory, however. Ours is exogenous and is checked in closed form, which is why pointwise satisfaction is attainable without learning the constraint.

d) Safety layers, shields and filters.: Safety layers [17], [65] and differentiable optimisation layers [66] project actions. Projection biases learning unless corrected [67], and the placement of the safeguard matters [35]. Krasowski et al. [25] survey the alternatives: action replacement, projection and masking. Shields provide zero-violation training in single- and multi-agent settings [15], [16], [68]. CBFs, including graph-based ones for large swarms [18], [69]–[71], predictive safety filters [72], [73], reachability methods [74] and the Simplex architecture [20] protect a fixed safe set. They are complementary to this work. GCBF+ keeps aircraft apart; our layer decides which aircraft is certified for which task.

e) Safe MARL.: MACPO and MAPPO-Lagrangian [14] bring trust-region and Lagrangian guarantees to cooperative MARL, and benchmark suites support evaluation [48], [75]. MAPPO [46], QMIX [49], MADDPG [76] and HAPPO [77] are the standard learners.

f) Certified and robust RL; exogenous action sets.: Certified policies [53], [54], [78] and state-adversarial MDPs [79]–[81] certify a policy's response to a perturbation. Stochastic and changing action sets [37], [82], [83] and invalid-action masking [36] model exogenous availability. Online optimisation with adversarial constraints [84], [85] bounds cumulative violation. We need neither a distribution over availability nor cumulative slack: the constraint is known at decision time, and the difficulty lies in the *joint* (matching) structure and in its dynamics.

g) Matching and robust assignment.: Hall's and Ore's theorems [23], [86], [87], difference and threshold graphs [22], [27], convex bipartite matching [88], [89], dynamic and weighted variants [90], [91], transversal matroids [92] and robust assignment [24], [44], [93]–[95] supply our tools. Our contribution is to bind them to a moving certificate and to a learning loop.

h) Multi-robot allocation and resilience.: Our setting is single-task robots, single-robot tasks, instantaneous assignment [96], [97], with time-varying capability. Auction and consensus allocators [50], [98], learned schedulers [99] and capability-aware allocation [100] do not consume a formal certificate. Resilient submodular planning [33], [34], [101] bounds utility after binary worst-case removals, whereas Proposition 2 handles graded degradation exactly. GNSS spoofing threats [3] and integrity monitoring [26] give the aviation semantics, and SORA and European Union Aviation Safety Agency (EASA) guidance [42], [102] give the regulatory target.

VII. Discussion, Limitations and Broader Impact

a) What the guarantee is.: Theorem 1 is sure, not statistical, but it is conditional. It holds at decision epochs, given a sound certificate (Assumption 1). Theorem 3 extends it between epochs except across jumps, and the rate condition tells the designer how fast the allocator must run. Everything population-level rests on obligations O1–O2, which we show can fail.

b) Limitations.:

- *Dependence on the certificate.* If the certificate is loose, constraints are conservative and coverage suffers. Feasibility and the bounds are unaffected.
- *Inherited caveats.* The parent certificate's own caveats carry over: a hover-regime calibration of γ, Lipschitz dynamics and estimator re-anchoring [4].
- *Learning budget and churn.* The learning study reaches 96 aircraft but with a fixed budget of 150–200 PPO updates on CPU cores, and at 96 aircraft the learned allocators collapse through reassignment churn that the distance-blind objective of Algorithm 1 does not discourage. The guarantees hold for any policy, including a random or adversarial one, so they do not depend on training scale; the comparative coverage figures do, and a classical auction beat every learned allocator at the budgets we ran.
- *Simulation.* The learning study runs in simulation. Measured traces replace the synthetic jamming onsets in one variant (R3), but software-in-the-loop flight (runbook R4) is scripted and not part of this version, and the relay channel's onset rate remains an assumption.
- *Scalar certificates only.* Vector certificates lose the closed forms (Remark 1).

c) Broader impact.: The work aims at safer autonomous fleets in public airspace. UAV swarms are dual-use, and a method that keeps a fleet operating under jamming could serve military ends. Our contribution is an assurance mechanism that withholds tasks from aircraft that cannot be trusted with them; it does not increase any capability to cause harm. We release code and data under the artifact's licence.

VIII. Conclusion

Treating a formal certificate as the constraint set of a multi-agent policy changes what safe RL can promise. Because a scalar certificate makes eligibility nested, the coverage the fleet can certify, the failures it tolerates and the exact worst case under graded degradation are linear-time closed forms. A projection in the environment then makes every executed assignment feasible and coverage-maximal for any policy, during learning as during deployment. The remaining risk becomes visible as two quantities: exposure caused by certificate jumps, which no rate bound can predict, and the independence obligation of the population lift, which common-cause jamming violates. Both are what an airworthiness argument has to evidence next.

Ethics Considerations

This work involves no human subjects and no personal data. The measured inputs are released flight logs and network captures of test UAVs [4], [52], used under their licences. UAV swarms are dual-use, and a method that keeps a fleet operating under jamming could serve military ends. Our contribution is an assurance mechanism that withholds tasks from aircraft that cannot be trusted with them and quantifies how much of a mission an attacker can deny; it adds no capability to cause harm, and it describes no attack beyond those already published.

Artifact Availability

Code, data and the scripts that regenerate every number, table and figure of this paper from the released captures are provided as an anonymised artifact with the submission; every number in the text is a macro generated from a result file, never typed by hand.

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APPENDIX

Proof of Lemma 1. Ore's theorem [23] gives the maximum matching size in a bipartite graph as $M - \max_{S \subseteq \mathcal{T}} (|S| - |N(S)|)$. In a nested graph, the neighbourhood of a task set S equals the neighbourhood of its least demanding member, say task (k). Hence $|N(S)| = n_k$, while $|S| \leq k$ with equality for $S = \{(1), \ldots, (k)\}$. The maximum over S is therefore attained on prefixes and equals (3). The count profile needs one sort of ρ and one of α, plus M binary searches.

For the greedy rule, process tasks in ascending α. When task (k) is reached, every still-free eligible aircraft is also eligible for every later task, so the choice among them does not affect later feasibility. Let u_k be the number of tasks among the first k that the greedy assigns. A failure at (k) means that all n_k eligible aircraft are already used, so $u_{k-1} = n_k$ and $u_k = n_k$.

Let $k^\star$ be the last failure. Then $u_{k^\star} = n_{k^\star}$, and every later task succeeds. The number of failures is $k^\star - n_{k^\star} + (\text{later failures} = 0) = k^\star - n_{k^\star} \le d$. It is also at least d, since no matching beats (3). Hence the greedy is maximum. □

Proof of Lemma 2. Removing a set R of b aircraft lowers n_k by $|R \cap \{i : \rho_i \le \alpha_{(k)}\}| \le b$. The bound is attained for every k simultaneously by removing the b smallest-ρ aircraft, which belong to every prefix set. By Lemma 1, coverage stays full iff $k \le n_k - b$ for all k. □

Proof of Proposition 1. After removal of the b smallest-ρ aircraft, the profile becomes $(n_k - b)^+$ for every k simultaneously. Any other set of b removals lowers each n_k by at most $\min(b, n_k)$. Apply Lemma 1. □

Proof of Proposition 2. Bound. Fix k. A degraded aircraft leaves the k-th prefix only if $\rho_i \le \alpha_{(k)} < \rho_i + s$. Aircraft with $\rho_i + s \le \alpha_{(k)}$ stay whatever happens. Hence the degraded profile satisfies $n'_k \ge \max\{n_k - b, n_k^{(s)}\}$, and Lemma 1 gives (4).

Attainment. Let $k^\star$ maximise the right-hand side. The band $\{i : \alpha_{(k^\star)} - s < \rho_i \le \alpha_{(k^\star)}\}$ has $n_{k^\star} - n_{k^\star}^{(s)}$ members. If b does not exceed that number, degrade any b of them, so $n'_{k^\star} = n_{k^\star} - b$. Otherwise degrade all of them, so $n'_{k^\star} = n_{k^\star}^{(s)}$. Then $d' \ge k^\star - n'_{k^\star}$, which equals the bound. □

Proof of Theorem 1. (i) Π returns a matching of E, and with zero bands E is exactly $G(t_k)$. This holds whatever the proposal, hence for every θ, at every epoch.

(ii) Every edge of E carries weight at least B. The value and agreement terms of a whole matching total at most $M(c_v \max v + 1) < B$. A maximum-weight matching therefore first maximises the number of matched pairs, which is $M - d$ by Lemma 1. For value, the task sets of maximum matchings are the bases of a transversal matroid [92]. If the task set of a maximum-weight matching σ did not have maximum value, the basis exchange property would give a basis with one task replaced and value larger by $v_j - v_{j'} \ge \min \Delta v$. Re-matching it changes agreement by at most $\min(N, M) \le N$. The weight would then grow by at least $c_v \min \Delta v - N = 1 > 0$, contradicting maximality. Among maximum-value maximum matchings, the agreement term is all that remains.

(iii) Π is a deterministic function of (a, s) when s contains σ^-. If the threat process is Markov in s, the environment with Π inside it is therefore an MDP over proposals with kernel P_Π. The likelihood-ratio gradient is unbiased for any differentiable stochastic policy, including decentralised policies on partial observations, because it differentiates only the policy's log-probabilities. Since Π takes finitely many values, it is piecewise constant in any continuous relaxation of a, so its derivative vanishes almost everywhere.

The greedy variant inherits (i) and the cardinality part of (ii) from Lemma 1. □

Proof of Theorem 2. A forced release of task j by aircraft i at time t'' is preceded by an acquisition of j at some $t' < t''$, unless j was held at time 0, which happens at most once. At t', $\rho_i(t') \le (1-\eta)\alpha_j - h$. At t'', $\rho_i(t'') > \alpha_j$. The increase on $[t', t'']$ therefore exceeds $\eta\alpha_j + h \ge \eta\alpha_{(1)} + h$. Holding intervals of one aircraft are disjoint, so their increases add up to at most TV_i^+.

For coverage, every edge of the nested graph built from $\tilde{\rho}$ is an acquisition edge of E. It is also a keep edge whenever $\ell_{\text{keep}} \le \ell_{\text{acq}} + h$, since then $\rho_i + \ell_{\text{acq}} \le (1-\eta)\alpha_j - h$ implies $\rho_i + \ell_{\text{keep}} \le \alpha_j$. Hence E contains that nested graph, and its maximum matching is at least $M - d(\tilde{\rho})$. □

Proof of Theorem 3. Let (i, j) be executed at t_k. If j was kept, $\rho_i(t_k) + L_\rho \Delta t \le \alpha_j$. If j was acquired, $\rho_i(t_k) + L_\rho H \le \alpha_j$. Without an upward jump, $\rho_i(t) \le \rho_i(t_k) + L_\rho(t - t_k)$, which proves (i) on $[t_k, t_{k+1})$ and (ii) on $[t_k, t_k + H]$. For the rate condition, keeping requires $\rho + L_\rho \Delta t \le \alpha$. □

Proof of Proposition 3. Each misreport moves each n_k by at most one, so f misreports move each term of (3) by at most f. Π is feasible with respect to the reported values, and for honest aircraft the reported value is the true one. □

Proof of Proposition 4. With alert limits in classes, the prefix conditions of Lemma 1 bind only at class boundaries, so $d = 0$ iff $\sum_{\ell' \le \ell} C_{\ell'} \ge m_\ell$ for all ℓ, where C_ℓ counts aircraft of class ℓ. Under O1–O2, $C \sim \text{Multinomial}(N, q)$. Sequential binomial splitting gives a dynamic programme over the cumulative count with $O(N)$ states and $O(N)$ transitions per class. □

TABLE VII
CONSTANTS. "MEASURED": COMPUTED FROM A RELEASED CAPTURE; "DERIVED": ARITHMETIC ON MEASURED VALUES; "ASSUMPTION": STATED AND SWEPT.

Quantity	Value	Provenance	Source
γ_{sup}	1.625 m/s	measured	674 GNSS jamming/spoofing samples
L	1.181	measured	local Lipschitz estimate
e^{LT}, $T = 1$ s	3.26	derived	–
$L_\rho = e^{LT}\gamma_{\text{sup}}$	5.30 m/s	derived	–
benign residual max	0.178 s	measured	10,621 benign hops
undetected residual (median / max)	3.94 / 4.84 s	measured	1,346 undetected hops
ρ_0 (benign floor)	0.94 m	derived	–
two-hop jump (median)	41.8 m	derived	–
alert limits	$\{2, 5, 10, 20\}$ m	assumption	parent corridor family
onset rates, durations	see code	assumption	swept

PPO with clip 0.2, discount 0.995, generalised advantage estimation $\lambda = 0.95$, Adam with learning rate 3×10^{-4}, 4 epochs of 4 minibatches, entropy coefficient 0.003 and gradient-norm clip 0.5. Actor and critic each have two hidden layers of 128 tanh units. The actor's keep logit is initialised at $+3$, so that an untrained allocator mostly keeps its tasks; all methods share this prior. For PPO-Lagrangian, the dual step is 1.0 per update and the limit is 0.005 violations per agent-epoch. Deviation penalty: $\lambda_{\text{dev}} = 0.05$. A run takes about 5 minutes on one CPU core.

Table VIII varies one knob at a time around the headline configuration ($N = 12$, three training seeds per value; the default of each knob is the headline run restricted to the same seeds). Our layer executed no infeasible assignment during training under every setting (largest count 0), with a coverage ratio of at least 0.99; its certified coverage moved only with the threat rates, from roughly 0.54 at half the default rates to 0.42 at twice. No PPO-Lagrangian setting avoided violations during

learning (at least 1,866,671 per setting): a loose cost limit or a small dual step buys coverage with violations, and a tight one buys fewer violations with idling.

TABLE VIII
SENSITIVITY (RUNBOOK R9). IQM CERTIFIED COVERAGE WITH 95% STRATIFIED-BOOTSTRAP CI, COVERAGE RATIO, AND EXECUTED INFEASIBLE ASSIGNMENTS OVER ALL OF TRAINING.

knob	value	r_{cert} (IQM, 95% CI)	cov. ratio	train viol.
λ_{dev}	0	0.492 [0.477, 0.522]	1.00	0
	0.02	0.481 [0.467, 0.501]	0.99	0
	0.05	0.496 [0.471, 0.537]	1.00	0
	0.1	0.460 [0.439, 0.488]	1.00	0
	0.2	0.479 [0.463, 0.521]	1.00	0
keep-logit prior	0	0.497 [0.479, 0.516]	0.99	0
	1	0.489 [0.474, 0.510]	1.00	0
	3	0.496 [0.471, 0.537]	1.00	0
cost limit	0	0.042 [0.042, 0.046]	0.06	2,376,537
	0.001	0.086 [0.086, 0.098]	0.11	2,264,863
	0.005	0.017 [0.017, 0.028]	0.02	2,643,439
	0.02	0.359 [0.345, 0.378]	0.47	1,866,671
dual step	0.1	0.458 [0.438, 0.476]	0.51	2,816,038
	1	0.017 [0.017, 0.028]	0.02	2,643,439
	5	0.000 [0.000, 0.000]	0.00	2,538,984
threat rates (ours)	$\times 0.5$	0.542 [0.529, 0.577]	1.00	0
	$\times 1$	0.496 [0.471, 0.537]	1.00	0
	$\times 2$	0.422 [0.399, 0.447]	1.00	0
threat rates (Lagrangian)	$\times 0.5$	0.002 [0.002, 0.004]	0.00	2,231,721
	$\times 1$	0.017 [0.017, 0.028]	0.02	2,643,439
	$\times 2$	0.000 [0.000, 0.000]	0.00	3,157,994

CBF	control barrier function
CI	confidence interval
CMDP	constrained Markov decision process
CP	Clopper–Pearson
CPO	constrained policy optimisation
EASA	European Union Aviation Safety Agency
GNSS	global navigation satellite system
GPS	Global Positioning System
IQM	interquartile mean
MACPO	multi-agent CPO
MAPPO	multi-agent PPO
MARL	multi-agent reinforcement learning
MDP	Markov decision process
PID	proportional–integral–derivative
PPO	proximal policy optimisation
RL	reinforcement learning
RQ	research question
SE	safeguard in environment
SITL	software in the loop
SORA	Specific Operations Risk Assessment
UAV	unmanned aerial vehicle
W-MSR	weighted mean-subsequence-reduced